MIRACLE: Multimodal Information Retrieval via a Combined In-Memory Processing and Content Addressable Memory Approach

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***Abstract*—The rapid advancement of information technology has brought multimodal information retrieval into the research spotlight. Neural networks, particularly Transformers, have emerged as the dominant solution for extracting multimodal feature vectors. While neural network acceleration has been extensively explored, the subsequent retrieval stage in multimodal scenarios remains under-optimized. Conventional retrieval approaches, such as cosine similarity sorting on von Neumann architectures, suffer from significant data migration and computational inefficiencies. Hashing methods enhance storage and computation efficiency but encounter challenges in energy-efficient implementation and mitigating accuracy losses due to modal heterogeneity. This paper presents a hybrid architecture that integrates in-memory processing (PIM) and content-addressable memory (CAM) to address these challenges. Transformer-extracted features are processed via in-memory random hashing leveraging device-intrinsic properties, with CAM facilitating parallel search space reduction. A final cosine similarity reranking stage refines the results while balancing accuracy with energy efficiency. Experimental evaluations validate that the proposed method, when compared to the baseline traditional CPU-based cosine similarity retrieval, 1) achieves almost identical level of accuracy, dramatically outperforming other pure CAM-based Hamming distance retrieval approaches; and 2) reduces latency by $9.45\times$ and energy consumption by $30.20\times$.**

***Index Terms*—multimodal information retrieval, hashing, in-memory processing, content-addressable memory**

I. Introduction

The rapid expansion of Internet usage has led to data being represented in diverse forms, collectively known as multimodal data. Efficient retrieval of relevant information from such heterogeneous and vast datasets has become increasingly challenging, necessitating the development of multimodal information retrieval systems capable of delivering comprehensive, multidimensional search results [1].

In multimodal information retrieval, the process begins with feature extraction, where neural networks are used to derive critical feature vectors from diverse data modalities. Transformers [2], with their powerful attention mechanisms and cross-modal adaptability, have emerged as one of the leading architectures for this purpose. The multimodal information retrieval process, typically modeled as an Approximate Nearest Neighbor (ANN) problem, involves identifying entries across modalities with high similarity to the query item. However, traditional methods, such as cosine similarity sorting, have significant energy and latency issues due to frequent data migrations between memory and processors, exacerbated by the von Neumann bottleneck [3].

Hashing methods address some of these challenges by mapping high-dimensional feature vectors into compact hash sequences while preserving similarity relationships [4]. The retrieval workflow in hash-based systems involves three phases: feature extraction, hash mapping, and feature retrieval. During the hash mapping phase, methods like locality-sensitive hashing (LSH) generate hash codes based on matrix-vector multiplications (MVM) with hyperplane normal vectors. However, these operations are computationally intensive, particularly in generating random normal vectors and performing the required MVM calculations. Secondly, hash-based retrieval relies on frequent data transfers for similarity computations and sorting. Thirdly, dimension reduction during hashing often degrades retrieval accuracy in multimodal scenarios.

Emerging technologies such as in-memory processing (PIM) [5]–[7] and content-addressable memory (CAM) [8] offer promising solutions to mitigate the von Neumann bottleneck by enabling in-memory operations for hash mapping and retrieval. Among different PIM technologies, spin-transfer-torque magnetic random access memory (STT-MRAM) stands out as an ideal platform due to its non-volatility, high endurance, and superior storage density [9], [10]. It can efficiently support energy-efficient matrix multiplications and random vector generation through its intrinsic resistive device properties.

CAM, on the other hand, facilitates high-speed parallel matching operations and hence is potentially well-suited for frequent retrieval tasks in multimodal systems. However, recent advancements utilizing CAM for hash sequence retrieval are limited to single-modal in areas such as image retrieval [11] and memory-augmented neural networks [3], [12]. For multi-modal contexts, shorter hash sequences compromise retrieval accuracy, while longer sequences exceed the matching capabilities of CAM.

In this paper, we propose MIRACLE: a Multimodal Information Retrieval Framework via a Combined In-Memory Processing and Content Addressable Memory Approach. MIRACLE integrates PIM's in-situ computation capabilities with

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CAM's parallel search characteristics, enhanced by a refined cosine similarity ranking stage where we use CAM as a filter to narrow down the search space before performing high-precise cosine similarity reranking. In addition to the novel framework of MIRACLE, we make the following contributions:

- We develop an energy-efficient hashing method to leverage STT-MRAM's intrinsic device properties for random vector generation and efficient matrix multiplications.
- We design a threshold-based filtering strategy that exploits CAM's parallel search capabilities to reduce the retrieval search space, improving both accuracy and energy efficiency.
- We implement MIRACLE and show that it maintains the baseline approach's retrieval accuracy while drastically reducing latency and energy consumption by $9.45\times$ and $30.2\times$, respectively.

The remainder of this paper is organized as follows: Section II presents the background knowledge on multimodal feature extraction, hashing method for the ANN problem, and PIM and CAM based on STT-MRAM. Section III describes the proposed hybrid architecture for multimodal information retrieval. Sections IV and V elaborate the implementation of locality-sensitive hash mapping in PIM and a CAM-based search space filtering, respectively. Section VI reports experimental results and Section VII concludes.

II. Preliminaries

A. Multimodal Feature Extraction Network

Different data modalities require unique feature extraction networks, with the Transformer [2] being prominent due to its effective attention mechanism, flexible tokenization, and universal architecture. It has been successfully applied in text, image, video and audio processing [13], [14], showcasing strong scalability in multimodal contexts. CLIP [15] represents a significant advancement in this area, utilizing two Transformer encoders—one for text and one for images. Through contrastive learning, CLIP establishes a shared representation space that maximizes cosine similarity between matched image-text pairs, thereby enhancing multimodal retrieval capabilities. In this work, we utilize CLIP for feature vector extraction from both image and text modalities.

B. Hashing Method

Hashing methods provide efficient solutions for approximate nearest neighbor search problems. These methods are categorized into data-independent locality-sensitive hashing (LSH) and data-dependent learning to hash approaches [4]. LSH has demonstrated widespread success in both academic research and industrial applications, from computer vision tasks to Google's sim-hash implementation for duplicate webpage detection.

For cosine similarity preservation, sign random projection hashing serves as the corresponding locality-sensitive hashing technique, which partitions the feature space using random hyperplanes. The LSH-generated hash sequence h for a high-dimensional vector x can be expressed as:

$$h = sgn(x \cdot N) \tag{1}$$

where sgn represents the sign function and N denotes a matrix of random zero-mean vectors. Ternary LSH (TLSH) addresses the boundary problem where similar vectors near hyperplanes may receive different hash codes by introducing a third state X, which maintains zero Hamming distance to both 0 and 1. Section IV details the implementation of LSH within PIM crossbars.

C. PIM and CAM based on STT-MRAM

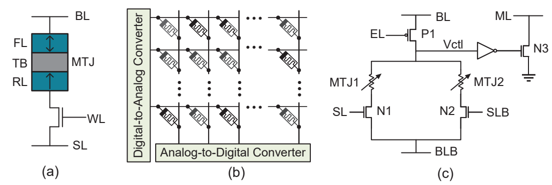

Fig. 1: PIM and CAM based on STT-MRAM: (a) STT-MRAM bitcell, (b) analog crossbar and (c) CAM cell.

Spin-transfer-torque magnetic random access memory (STT-MRAM) represents an advanced non-volatile memory technology utilizing Magnetic Tunnel Junctions (MTJs) for data storage. As illustrated in Fig. 1 (a), each MTJ consists of reference and free ferromagnetic layers, with their relative resistance indicating the logic state. STT-MRAM functions by passing a spin-polarized current to alter the magnetic direction of the free layer, enabling smaller dimensions and enhanced immunity to stray magnetic fields. Read and write operations are conducted by applying voltage across selected SLs and BLs.

Analog in-memory processing exploits resistive memory devices' inherent characteristics for computational operations, particularly matrix-vector multiplications (MVMs). As shown in Fig. 1 (b), matrix elements are encoded as conductances within the array, with input vectors applied as row voltages. Column currents represent MVM results according to Kirchhoff's law, supported by peripheral Digital-to-Analog Converters (DACs) and Analog-to-Digital Converters (ADCs).

CAM facilitates in-memory parallel retrieval of hash sequences with O(1) time complexity, offering dual advantages of accelerated retrieval and elimination of processor-memory data transfer overhead. The structure of the CAM cell based on STT-MRAM is illustrated in Fig. 1 (c). A pair of MTJs encodes the stored values of 0/1 or X through their resistance, while a pair of input voltages (SL/SLB) controls which branch is connected to ground. The left-side PMOS and MTJ create a voltage divider affecting V_{ctl}, which controls the right-side NMOS switching. This mechanism produces distinct match line (ML) discharge patterns for matched versus unmatched entries. Multiple cells can be connected in parallel for word-level matching operations. Threshold filtering implements Hamming distance-based entry selection through comparator-based discharge speed monitoring at the match line terminus.

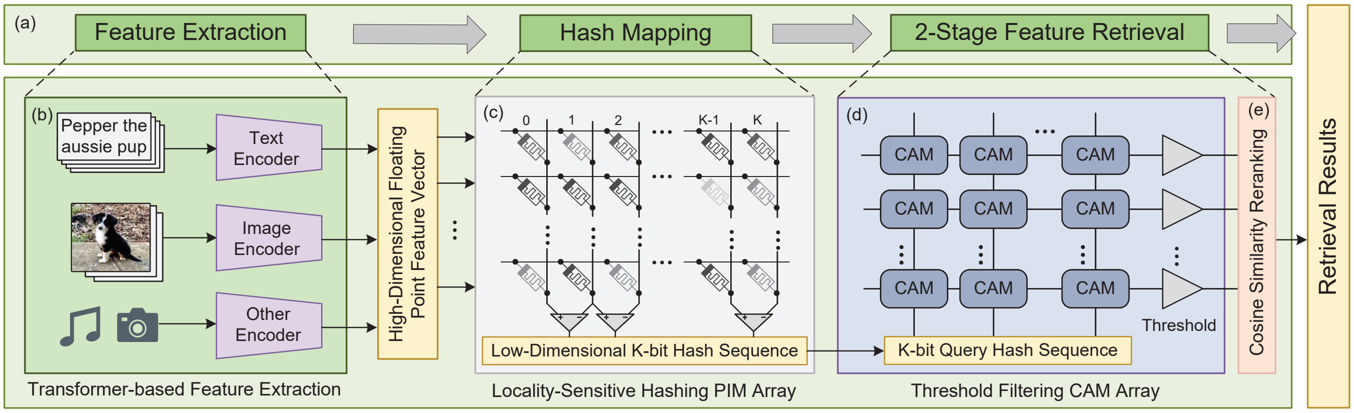

Fig. 2: Overall architecture: (a) workflow of multimodal information retrieval, (b) feature extraction using CLIP, (c) locality-sensitive hashing with analog crossbar, (d) filtering with CAM, and (e) cosine similarity reranking on subset filtered by CAM.

III. Overall Architecture

Hashing-based multimodal information retrieval typically involves three key phases: feature extraction, hash mapping, and feature retrieval.

The first phase is feature extraction, where neural networks extract critical feature vectors. Transformers have gained widespread adoption in multimodal feature extraction due to their powerful attention mechanisms, flexible token representations, and cross-modal universality. Transformer inference involves extensive matrix-vector multiplications, and its in-memory accelerators have been extensively researched.

In the hash mapping phase, high-dimensional feature vectors are compressed into compact hash codes using locality-sensitive hashing (LSH). This technique maps vectors with high cosine similarity to small Hamming distances, reducing storage and computational overhead. However, maintaining accuracy across multiple modalities and achieving energy-efficient hardware implementation remain significant challenges.

The feature retrieval phase computes hash sequence similarities to identify relevant items. While CAM enables rapid, parallel matching, its ability to discriminate similarity degrees diminishes with longer hash sequences.

To overcome these limitations, we propose a hybrid PIM-CAM retrieval architecture, as demonstrated in Fig. 2.

1) **Feature extraction:** A Transformer-based CLIP model extracts high-dimensional feature vectors.
2) **Hash mapping:** Device-level randomness in the PIM array enables energy-efficient locality-sensitive hashing, transforming floating-point vectors into 512-bit hash codes while preserving their similarity relationships.
3) **2-Stage Feature Retrieval:** First is the search space reduction stage. CAM's in-situ search capability filters entries based on a Hamming distance threshold, dramatically reducing the search space. To handle longer hash sequences, a segmented threshold matching and score aggregation mechanism is introduced. Next is the final refinement. Cosine similarity reranking is applied to the CAM-filtered subset to ensure high retrieval accuracy.

This hybrid architecture leverages PIM for computationally efficient hashing and CAM for rapid, parallel search, achieving a balance between performance, energy efficiency, and retrieval accuracy.

IV. PIM for Locality-Sensitive Hashing

A. Motivation and Challenge

Locality-sensitive hashing (LSH), as defined in Eq. (1), introduces several challenges when implemented in hardware. The first challenge is the significant data migration overhead associated with matrix-vector multiplication in traditional von Neumann architectures, which limits computational efficiency. Analog in-memory crossbars address this issue by encoding vectors in device conductance and input voltage, enabling dot product computation through current summation based on Kirchhoff's current law.

Two additional challenges emerge in adapting LSH to analog crossbars: First, efficiently storing random vectors with a mean of zero, as required by LSH principles. Second, extracting the sign of dot product results in an energy-efficient manner without relying on energy-intensive analog-to-digital converters (ADCs). Below demonstrates how to leverage the intrinsic hardware randomness of MTJ device in STT-MRAM and the differential encoding strategies to address the issues.

B. Leveraging Hardware Randomness for Random Vector Storage

The manufacturing process of MTJs introduces process variations, such as deviations in the thickness of the oxide barrier layer (γ_{tox}), the thickness of the free layer (γ_{tf}), and tunnel magnetoresistance (TMR) ratio variations, resulting in stochastic fluctuations in device resistance. These natural variations enable the generation of random vectors, which are essential for LSH.

To encode random vectors in an analog crossbar, the conductance of resistive devices represents individual vector elements. However, because device conductance is inherently positive, directly encoding zero-mean random vectors is infeasible. Differential encoding overcomes this limitation by representing a random vector as the difference in conductance between adjacent columns in the array. Specifically, each MTJ is initialized to a high-resistance state, producing conductance values that are random yet stable. The conductance difference between adjacent columns encodes a zero-mean random vector. The dot product between an input feature vector x and the encoded random vector a can then be expressed as:

$$\begin{aligned} x \cdot a &= \sum (V_i \cdot (G_i^+ - G_i^-)) \\ &= \sum (V_i \cdot G_i^+ - V_i \cdot G_i^-) \\ &= I^+ - I^- \end{aligned} \tag{2}$$

where input voltage V_i and conductance difference $(G_i^+ - G_i^-)$ between adjacent columns represent the values of the i-th dimension of x and a respectively. Therefore, I^+ and I^- represent the cumulative currents flowing through adjacent columns, effectively capturing the dot product result in the analog domain.

C. Energy-Efficient TLSH Hash Code Generation

Traditional analog computing systems rely on ADCs to digitize analog signals, introducing significant energy overhead. However, LSH requires only the sign of the dot product rather than its precise value. This allows direct conversion of analog signals to hash codes by comparing current magnitudes between adjacent columns, eliminating the need for ADCs. In this approach, the sign of the current difference determines the hash bit (0 or 1), while small differences are encoded as an "X" state to indicate ambiguity. This ternary logic provides flexibility while reducing computational complexity.

Fig. 3 illustrates the energy-efficient TLSH sensing circuit, which eliminates ADC usage by leveraging lightweight analog components. The circuit operates as follows: The circuit employs trans-impedance amplifiers (TIAs) to convert column current sums to corresponding voltage signals. Subtractors generate voltage differences between adjacent columns, reflecting the relative magnitudes of current sums. In the TLSH sensing block, comparators (Comp1, Comp2, and Comp3) process these voltage differences to produce 2-bit current-represented hash codes. Specifically, Comp1 determines the sign of the difference (positive/negative). Comp2 and Comp3 classify differences as small (ambiguous, "X") or significant (definitive, 0 or 1) based on a configurable threshold. The ternary encoding block outputs voltage levels representing hash codes: 0, 1, or grounded (for "X"). By directly converting current differences into hash codes, the TLSH sensing circuit minimizes energy consumption while maintaining high accuracy in hash code generation.

V. CAM FOR SEARCH SPACE FILTERING

A. Design Motivation

In MRAM-based CAM design, each hash code is represented by the conductance of two MTJs, while the query hash code is encoded in dual voltage. This configuration seamlessly integrates the 2-bit voltage-format hash codes generated by the

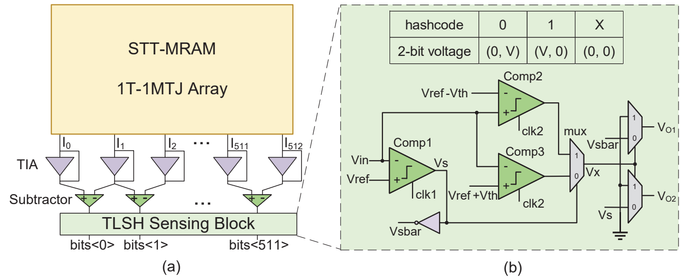

Fig. 3: The peripheral circuit for hashing mapping: (a) overall structure, and (b) TLSH sensing block that converts the input voltage difference into a 2-bit voltage representation of the hash code.

TLSH sensing block as CAM query voltages, enabling simultaneous hash mapping and Hamming distance computation in an energy-efficient manner.

However, deploying CAM for Hamming distance-based retrieval in multimodal scenarios reveals a critical trade-off. While high retrieval accuracy requires long hash sequences, the relatively low tunnel magnetoresistance (TMR) ratio of MRAM devices limits CAM's ability to differentiate similarity levels across extended sequences. This limitation diminishes retrieval precision for longer hash codes.

To address this, we repurpose CAM as a filtering mechanism rather than a comprehensive Hamming distance sorting solution. Specifically, CAM identifies entries with Hamming distances below predefined thresholds, narrowing the search space for subsequent cosine similarity reranking. To complement this filtering step, we propose a segment-based aggregation and concatenation mechanism that uses threshold-based scoring for individual hash code segments. This hybrid two-stage approach reduces latency and energy consumption while preserving high retrieval accuracy. The following sections detail the segmentation and merging process, as well as our methodology for selecting optimal thresholds.

B. CAM-Based Segmentation and Merging

Our proposed threshold filtering approach segments a K-bit hash sequence into n equal parts, each processed in parallel by a CAM sub-array of word length w. Fig. 4 illustrates the architecture.

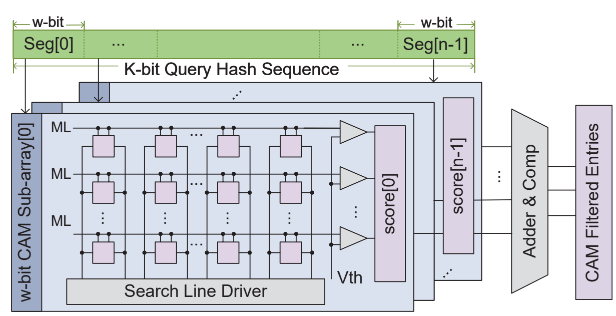

Fig. 4: Segmented threshold matching and merging mechanism of CAM filtering.

To optimize the trade-off between retrieval precision and reduced search space, we introduce a multi-threshold Hamming distance mechanism for each w-bit CAM sub-array. The scoring mechanism assigns a value to the i-th segment based on its Hamming distance relative to m predefined thresholds, as shown below:

$$score(i) = \sum_{j=0}^{m-1} (hamming\_dist(i) \leq threshold(j)) \quad (3)$$

For each stored entry, a total score is computed by summing the scores across all segments, as defined in Eq. (4):

$$total\_score(entry) = \sum_{i=0}^{n-1} score(i) \quad (4)$$

Entries exceeding a specified threshold are advanced to the next stage for further processing.

In our design, 512-bit hash sequences are divided into eight 64-bit segments, with each segment processed by a dedicated 64-bit CAM sub-array. We employ three distinct thresholds to evaluate Hamming distances within each segment. These thresholds are carefully calibrated to strike a balance between retrieval accuracy and search space reduction.

C. Threshold Selection

Given the fact that intra-class Hamming distances are generally smaller than inter-class distances, we defined a primary threshold th positioned at the boundary where intra-class distances predominantly fall below, while inter-class distances exceed this value.

To further optimize retrieval, two additional thresholds are introduced: A lower threshold (th_{low}) below th, which rewards intra-class pairs with very small Hamming distances, ensuring these entries achieve higher scores than non-matching pairs; A higher threshold (th_{high}) slightly above th, which allows more entries to be included in scoring to enhance overall retrieval accuracy. These thresholds are uniformly distributed across n segments, enabling segment-wise Hamming distance evaluation.

After implementing the segmentation and score aggregation mechanism, we analyzed the total score distributions of intra-class and inter-class pairs. A score threshold was then selected to effectively separate matching pairs (which predominantly score above the threshold) from non-matching pairs (which generally score below). Entries meeting or exceeding this threshold proceed to the next stage for cosine similarity reranking.

VI. Evaluations

A. Experimental Setup

Our evaluation framework encompasses multiple levels of analysis to assess both retrieval accuracy and hardware efficiency. At the device level, we employ the Brinkman model and Landau-Lifshitz-Gilbert (LLG) equation [10] to characterize MTJ behavior. The key parameters for MTJ simulation are demonstrated in TABLE I. Circuit-level analysis utilizes the Cadence suite for detailed simulation of PIM and CAM components. At the behavioral level, we implement Python-based simulations of hardware hash mapping and retrieval processes to evaluate retrieval accuracy.

TABLE I: Key Parameters for MTJ Simulation.

MTJ of STT-MRAM Parameters	Value
Effective Magnetic Anisotropy (H_K)	$3.58e5\ A/m$
Saturation magnetization (M_s)	$6e5\ A/m$
Gyromagnetic ratio	$1.76e7\ rad/(s \cdot T)$
Oxide Layer Thickness (t_{ox})	$0.85nm$
Damping Constant (α)	0.03
Tunnel Magnetoresistance (TMR)	3.6

For fair comparison, we utilize the ViT-L/14@336px model from [15] for feature extraction, mapping 768-dimensional float16 feature vectors to 512-bit hash sequences. We evaluate our method on three widely-used multimodal information retrieval datasets: MSCOCO [16], Flickr8K [17], and Flickr30K [18]. Following the CLIP's evaluation protocol, we employ recall rate (R@k) as the primary accuracy metric.

To provide a comprehensive analysis, we compare both accuracy and hardware efficiency. Accuracy evaluations include comparisons against CPU-based cosine similarity retrieval and hash-based Hamming distance retrieval (both CPU and in-memory implementations). Hardware efficiency metrics, including latency and energy consumption, are benchmarked against PyTorch-based similarity calculation and sorting operations on an Intel(R) Core(TM) i9-14900KF CPU, providing a standardized baseline against conventional computing approaches.

B. Optimal Threshold Selection

We demonstrate the proposed threshold selection approach with the Flickr8K dataset. Fig. 5 (a) reveals distinct Hamming distance distributions: intra-class distances predominantly typically fall below 220, while inter-class distances are generally above this value. Based on this natural separation, 220 is selected as the primary threshold. To refine discrimination, we introduce additional threshold at 210 to reward smaller intra-class distances, and a threshold at 225 to ensure comprehensive coverage.

After aggregating scores across eight segments, Fig. 5 (b) and (c) illustrate the resulting intra-class and inter-class score distributions. Intra-class entries consistently achieve scores of 9 or higher, while inter-class entries cluster below this value. Consequently, we adopt 9 as our score threshold, filtering entries scoring 9 or above into the candidate subset.

Fig. 5 (d) explores the trade-off between search space size and retrieval accuracy under varying thresholds. Lower thresholds expand the filtered set, increasing reranking overhead but improving recall rates. Retrieval accuracy improves rapidly with increasing search space before plateauing. Balancing retrieval accuracy with energy consumption, we select thresholds that filtered approximately 13.9% of the search space.

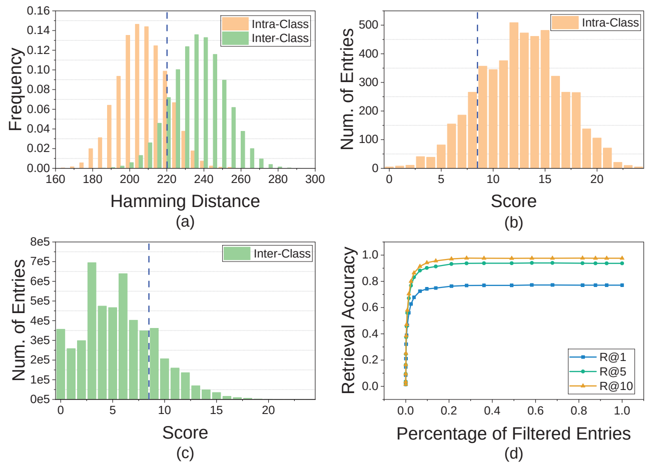

Fig. 5: Threshold selection criteria: (a) distribution of Hamming distances for intra-class and inter-class pairs, (b) distribution of total scores for intra-class pairs, (c) distribution of scores for inter-class pairs, and (d) trade-off between retrieval accuracy and filtered entry ratio across different threshold values.

C. Retrieval Accuracy

For a comprehensive evaluation, we compare recall rates across three approaches: traditional CPU-based cosine similarity retrieval [15], full in-memory hash and Hamming distance retrieval using PIM and CAM [12], and our proposed method. To identify the primary sources of accuracy loss, we also evaluate how different implementations of hash and Hamming distance retrieval affect recall rates, as shown in Fig. 6.

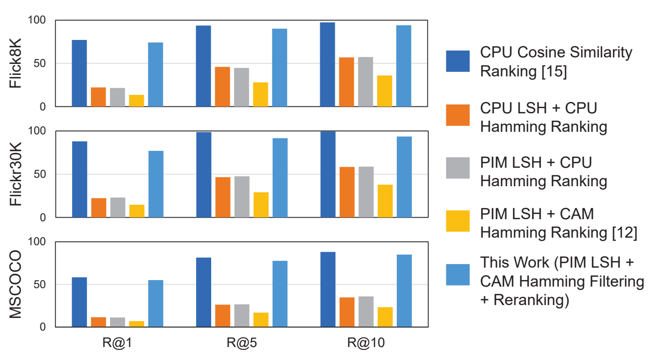

Fig. 6: Recall rate comparison between different implementations of hashing and retrieval.

In multimodal contexts, Hamming distance retrieval on LSH-generated hash sequences shows significant accuracy degradation compared to cosine similarity retrieval, with CAM's matching limitations further exacerbating this loss. While in-memory LSH implementation maintains accuracy comparable to CPU-based approaches, PIM delivers superior energy efficiency benefits. Our approach preserves strong recall rates, with only a $1\% \sim 4\%$ degradation. Notably, despite the MSCOCO dataset containing $25,000$ entries, the recall rate degradation is only 3%, demonstrating the scalability of our approach for large-scale retrieval tasks.

D. Latency and Energy Consumption

CAM-based Hamming distance ranking exhibits Unacceptable low accuracy in multimodal contexts, rendering it impractical for real-world applications. Therefore, we restrict our latency and energy consumption comparisons to CPU-based cosine similarity ranking.

TABLE II compares our proposed method against CPU ranking, examining CAM-filtered entries, latency, and energy consumption. Compared to CPU-based cosine similarity computation and ranking, the hash mapping and CAM filtering account for only a small fraction of the total energy consumption and latency, thanks to the energy-saving in-memory computation and search capabilities of MRAM-based PIM and CAM. Our CAM filtering approach achieves an average $10.85\times$ reduction in cosine similarity search entries, resulting in on average $9.45\times$ lower latency and $30.20\times$ lower energy consumption.

TABLE II: Reduction Ratios in Filtered Search Space, Latency and Energy Consumption Compared to the CPU Baseline.

Dataset	Num. of Entries	Latency	Energy
Flickr8K	10.23	5.45	6.58
Flickr30K	13.80	6.28	7.83
MSCOCO	8.53	16.62	76.20
Average	10.85	9.45	30.20

Note that the efficiency gains scale with dataset size due to the O(nlogn) complexity of CPU-based similarity sorting. For instance, filtering Flickr8K's search space from $5,000$ to 362 entries yields a $7.83\times$ reduction in energy consumption, while filtering MSCOCO from $25,000$ to $2,930$ entries achieves a $76.20\times$ reduction. These results demonstrate our method's significant advantages in large-scale retrieval applications, where the benefits become increasingly pronounced with larger datasets.

VII. Conclusion

This paper presents a novel multimodal information retrieval architecture that integrates PIM and CAM, achieving both high retrieval accuracy and energy efficiency. In the feature extraction stage, the Transformer model is employed to extract multimodal feature vectors. In the hash mapping stage, device-level randomness is leveraged to implement in-memory hashing mapping. In the feature retrieval stage, CAM filters candidate subsets based on Hamming distance thresholds, followed by precise cosine similarity retrieval within the reduced search space. The accuracy has significantly improved compared to pure Hamming distance sorting. When compared to cosine similarity ranking, our method achieves almost identical level of accuracy with a $9.45\times$ reduction in latency and $30.20\times$ reduction in energy consumption. Furthermore, our method exhibits excellent scalability in handling large-scale data, which is of significant practical importance given the exponential growth of multimodal data nowadays.

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